

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Probability

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the probability of getting a prime number on a die.

Answer:

Probability = favourable/total = $3/6 = 1/2$.

Question 2

Find the probability of getting an even number greater than 3 on a die.

Answer:

Probability = favourable/total = $2/6 = 1/3$.

Question 3

Find the probability of getting a heart from a deck.

Answer:

Probability = favourable/total = $13/52 = 1/4$.

Question 4

Find the probability of getting an ace from a deck.

Answer:

Probability = favourable/total = $4/52 = 1/13$.

Question 5

Find the probability of getting a ball that is not blue when 3 of 12 are blue.

Answer:

Probability = favourable/total = $9/12 = 3/4$.

Question 6

Find the probability of getting a red ball when 4 of 10 are red.

Answer:

Probability = favourable/total = $4/10 = 2/5$.

Question 7

Find the probability of getting a multiple of 4 from 1 to 20.

Answer:

Probability = favourable/total = $5/20 = 1/4$.

Question 8

Find the probability of getting a multiple of 3 from 1 to 30.

Answer:

Probability = favourable/total = $10/30 = 1/3$.

Question 9

Find the probability of getting a head when one fair coin is tossed.

Answer:

Probability = favourable/total = $1/2 = 1/2$.

Probability - Continued

Question 10

Find the probability of getting a square number from 1 to 16.

Answer:

Probability = favourable/total = $4/16 = 1/4$.

Question 11

A bag has 4 red, 3 blue and 5 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 9; total = 12. Probability = $9/12$.

Question 12

A bag has 5 red, 2 blue and 3 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 8; total = 10. Probability = $8/10$.

Question 13

A bag has 6 red, 4 blue and 2 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 8; total = 12. Probability = $8/12$.

Question 14

A bag has 3 red, 5 blue and 4 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 7; total = 12. Probability = $7/12$.

Question 15

A bag has 8 red, 1 blue and 7 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 15; total = 16. Probability = $15/16$.

Question 16

A bag has 2 red, 6 blue and 4 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 6; total = 12. Probability = $6/12$.

Question 17

A bag has 7 red, 3 blue and 5 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 12; total = 15. Probability = $12/15$.

Question 18

A bag has 9 red, 2 blue and 1 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 10; total = 12. Probability = $10/12$.

Question 19

A bag has 4 red, 4 blue and 4 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 8; total = 12. Probability = $8/12$.

Probability - Continued

Question 20

A bag has 10 red, 5 blue and 5 green balls. Find the probability of not picking blue.

Answer:

Not blue balls = 15; total = 20. Probability = $15/20$.

Question 21

State the main formula, rule or theorem used in Probability.

Answer:

$P(E) = \text{number of favourable outcomes} / \text{total equally likely outcomes}$.

Question 22

What do the important symbols or quantities in Probability represent?

Answer:

An event is a set of outcomes; $0 \leq P(E) \leq 1$.

Question 23

Write the correct method for solving a Probability numerical.

Answer:

List the equally likely outcomes, count favourable outcomes, divide by total and simplify the fraction.

Question 24

Which unit, sign convention or condition must be checked in a Probability answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Probability be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Probability questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Probability numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Probability?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Probability - Continued

Question 29

How can a student decide which formula to use in Probability?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Probability?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY